# CWR in situ · Installation & user guide

Field recording of crop wild relatives (CWR) — international version

*How to install the app on a phone (Android or iPhone), record a population in the field, reuse genetic-reserve polygons, and protect your data. Read it fully before your first field trip.*

## Part 1 · Installing the app

CWR in situ is a web application installed from a web address. You (or your coordinator) will host it and share a **URL** of the form https://txiriondo.github.io/CWR-in-situ/FlorCWR.html. The procedure differs by phone; follow only your section.

*Host this English version at its own URL, separate from any Spanish version. Two web apps served from the same path would overwrite each other's offline cache and stored records.*

### A. Android (Samsung, Xiaomi, Google Pixel, etc.)

1. Open the URL in **Google Chrome**.
2. Wait for the banner **"✓ available offline"**.
3. Open the Chrome menu (**three dots ⋮**) and choose **"Add to Home screen"** / **"Install app"**.
4. Open the app from the new icon and grant the **location** permission when asked.

### B. iPhone (iOS)

**Use Safari, not Chrome.** The install and storage features only behave correctly in Safari.

1. Open the URL in **Safari** and wait for **"✓ available offline"**.
2. Tap the **Share** button (square with an upward arrow) and choose **"Add to Home Screen"**.
3. Open the app from the icon and grant the **location** permission.

### Offline base map (both systems)

The app always starts offline, but the **map background** must have been loaded earlier with a connection. Before the field, with coverage, open the map and pan/zoom over your working zone so the tiles are cached for offline use.

### Automatic updates

When a new version is published, the app shows an **"A new version is available — Update"** bar; tap **Update**. The first time, or after a service-worker change, you may need to force it once (hard reload, or clear site data after exporting a backup).

## Part 2 · Recording a population

The form is ordered to match field workflow. It **starts with the genetic reserve**, then the population details.

### Reserve block (top of the form)

- **Locality** — place name / site / municipality.
- **Genetic reserve name** — free text, e.g. "Grazalema 1".
- **Saved reserves** — a dropdown (appears once you have saved at least one reserve) to reuse a reserve already recorded; see below.
- **Reserve area** — the reserve polygon, on the map.

### Drawing the reserve polygon

Tap "Start polygon", then build the outline in any of three ways:

1. **Tap the map** to place vertices by eye.
2. **Add GPS vertex**: stand at each corner and tap.
3. **Track on foot**: auto-record a point every few seconds while you walk the perimeter.

Use **Undo** and **Close polygon**. The area is computed in m²/ha. Tap **"Follow my position"** to show a live blue dot of your current location on the map (it stays visible even with the frequency grid on).

## Reusable genetic reserves

When a **reserve name is filled in and the polygon is closed**, the reserve is **saved automatically** (name + polygon + locality) in a store separate from the population records. In a later record, pick it from **Saved reserves** and tap **Load**: the polygon, the reserve name and the locality are restored, so you don't re-survey them. The occupancy-frequency grid is **per population**, so it resets on load — define it for the new record if needed.

## Population details

- **Species** (free text) and optional **Family**. If a national checklist is loaded (see below), the Species field suggests names as you type; you can still enter a name that is not on the list.
- **ID certainty, Observation date** (required), **Point coordinates** (GPS or typed; UTM ETRS89 shown).
- **Locality** is already filled from the reserve; **Habitat (EUNIS), Substrate, Phenology, Cover, Herbarium voucher, Photos, Recorder, Notes** as applicable.

## Species checklist (optional)

The international version has **no built-in species list**. Each country can load its own CWR checklist so the **Species** field offers autocomplete. In the **Data** tab, under **"Species checklist"**, load a file with one scientific name per line (`CSV` or `TXT`; a header row is detected automatically) or a `JSON` array of names. A counter shows how many taxa are loaded; **"Remove checklist"** returns to free-text mode. The checklist is stored on the device and travels inside the JSON backup, so a coordinator can prepare it once and share it with the team.

*The checklist only assists typing — it never restricts what you can record. A name absent from the list is still saved, so unexpected finds are never blocked.*

## Population size estimation

| Method | What it does |
|---|---|
| **Total census** | Type the Abundance directly and choose any Count unit. |
| **Transect** | Enter transect size (m²) and number of individuals. Abundance = individuals × reserve-area / transect size. Count unit forced to "Individual". |
| **Frequency** | Lay a UTM grid over the polygon; mark occupied cells (GPS or tap). Abundance = occupancy frequency (%). Count unit forced to "Frequency (%)". |

*Transect note: assumes the transect density is representative of the whole reserve area (homogeneous distribution). For aggregated populations, several transects and an average are more reliable.*

## Occupancy frequency grid

Choose a cell size (5, 10, 25, 50, 100 m or 1 km). The grid is clipped to the polygon. Walk the population and tap **"Mark presence (GPS)"** at each occupied point; frequency = occupied cells × 100 / total cells.

*Accuracy: a phone GPS is typically ±4–10 m, so 5–10 m cells fall below GPS accuracy; the app warns when accuracy exceeds half the cell size. Confirm fine-grid cells visually or by tapping.*

# Part 3 · Exporting and backing up data

Records are stored **only in the browser of the phone that took them**. The **Data** tab offers **CSV** (flat table), **GeoJSON** (points + polygons for QGIS; WGS84 with UTM ETRS89 as attributes) and **JSON backup** (full state, the only re-importable format — it also includes your saved reserves and any loaded checklist). Field names are in English.

## Backup routine

**After each field day, before closing the app, export a JSON backup and upload it to your shared repository.**

> **iPhone users: the backup is critical.**
> Safari may erase a web app's stored data after ~7 days of inactivity. Export the JSON backup after every outing, and open the app every few days even if you record nothing.

File-name convention: `florcwr_YYYYMMDD_zone_initials.json`. Where several people record together, one person merges the individual backups in a dedicated browser, checking duplicates by species and nearby coordinates.

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*While the app has no shared back-end database, this manual routine keeps the team's work from being lost or overwritten. Saved reserves are shared between records on the same device, but not automatically between phones — they travel inside the JSON backup.*